INCH-POUND

MIL-C-39003/6E
13 November 1995
SUPERSEDING
MIL-C-39003/6D
31 January 1992

MILITARY SPECIFICATION SHEET

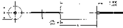
CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILITY, STYLE CSR23

FAILURE RATE LEVELS M, P, AND R ARE INACTIVE FOR NEW DESIGN AFTER 2 JANUARY 1996.

FAILURE RATE LEVEL S IS INACTIVE FOR NEW DESIGN AFTER 30 AUGUST 1995.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39003.



Case size	Dimensions L ±.031	D ±.016	H ±.002	J max
A	.286	.135	.020	.422
B	.474	.185	.020	.510
C	.686	.289	.025	.822
D	.786	.351	.025	.922

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.135	3.43	.474	12.04
.015	0.38	.185	4.70	.510	12.95
.016	0.41	.250	6.35	.686	17.42
.020	0.51	.286	7.26	.786	19.96
.025	0.64	.289	7.34	.822	20.88
.031	0.79	.351	8.92	.922	23.42
.094	2.39	.422	10.72	1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The case insulation shall extend .015 (0.38 mm) beyond the end of the case. [illegible]
4. Units shall be supplied [illegible] for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configuration.

Ⓒ Denotes changes.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: [illegible], hermetically sealed.

Material: [illegible] (see [illegible]).

Type of seal: [illegible].

(C) Leads: Types [illegible], [illegible], [illegible], or [illegible] of MIL-STD-1276.

DC rated voltage: See table I. Above 85°C voltage derating is required (see [illegible]).

DC leakage (DCL): See table I.

Capacitance (Cap.) (nom): See table I.

Cap. tolerance: See table I.

Dissipation factor (DF): See table I.

Operating temperature: -55°C to +125°C.

Failure rate level: See table I.

Stability at low and high temperatures: In accordance with MIL-C-39003.

Step 1 (+25°C):

DCL: See table I.

Cap.: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Cap.: Shall not change more than 10 percent of step 1 measured value.

DF: See table I.

Step 3 (+25°C):

DCL: See table I.

Cap.: Shall not change more than [illegible] percent of step 1 measured value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

Cap.: Shall not change more than 10 percent of step 1 measured value.

DF: See table I.

Step 5 (+125°C):

DF: See table 1.

Cap.: Shall not change more than ±12 percent of step 1 measured value.

IR: See table 1.

Step 6 (+25°C):

DF: See table 1.

Cap.: Shall not change more than ±2 percent of step 1 measured value.

IR: See table 1.

Surge voltage: In accordance with MIL-C-39003.

DCL: See table 1.

Cap.: Shall not change more than ±5 percent of initial measured value.

DF: See table 1.

Life: Method 108 of MIL-STD-202.

250, 1,000, and 2,000 hours.

At +25°C (as applicable):

DCL: See note 2.

Ⓒ ΔCap.: Shall not change more than ±5 percent of initial measured value.

DF: See table 1.

At +85°C:

DCL: See table 1.

At +125°C:

DCL: See table 1.

Ⓒ 4,000, 6,000, 8,000, and 10,000 hours (Appendix only):

At +25°C (as applicable):

DCL: 200 percent of initial limit.

Cap.: Shall not change more than ±10 percent of initial measured value.

DF: See table 1.

At +85°C:

DCL: 300 percent of initial limit.

At +125°C:

DCL: 200 percent of initial limit.

Moisture: In accordance with MIL-C-39003.

Part or identifying number (PIN): M39003/09-(dash number from table 1).

TABLE I. [illegible]

TABLE I. [illegible] Continued.

Rated voltage	Capacitance	Capacitance tolerance	DC leakage at: +25°C	DC leakage at: +85°C	DC leakage at: +125°C	Dissipation factor -55°C 25°C	Dissipation factor +85°C +125°C	Case size	M39003/03- Failure rate level for (%/1,000 hours): M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	µA	percent	percent								
15	6.8	10	1.3	13.0	16.3	4	4	A	0132	0232	0332	0432	2032	3032	4032
15	6.8	20	1.3	13.0	16.3	4	4	A	0133	0233	0333	0433	2033	3033	4033
15	8.2	10	1.5	15.0	18.8	4	4	A	0134	0234	0334	0434	2034	3034	4034
15	56.0	10	6.0	60.0	75.0	6	6	B	0135	0235	0335	0435	2035	3035	4035
15	33.0	20	6.0	60.0	75.0	6	6	B	0136	0236	0336	0436	2036	3036	4036
15	82.0	10	6.0	60.0	75.0	6	6	B	0137	0237	0337	0437	2037	3037	4037
15	150.0	10	15.0	150.0	188.0	8	8	C	0138	0238	0338	0438	2038	3038	4038
15	150.0	20	15.0	150.0	188.0	8	8	C	0139	0239	0339	0439	2039	3039	4039
15	180.0	10	15.0	150.0	188.0	8	8	C	0140	0240	0340	0440	2040	3040	4040
15	220.0	10	20.0	200.0	250.0	8	8	D	0141	0241	0341	0441	2041	3041	4041
15	220.0	20	20.0	200.0	250.0	8	8	D	0142	0242	0342	0442	2042	3042	4042
15	270.0	10	20.0	200.0	250.0	8	8	D	0143	0243	0343	0443	2043	3043	4043
15	330.0	10	20.0	200.0	250.0	8	8	D	0144	0244	0344	0444	2044	3044	4044
15	330.0	20	20.0	200.0	250.0	8	8	D	0145	0245	0345	0445	2045	3045	4045
20	4.7	10	1.0	8.0	10.0	4	4	A	0146	0246	0346	0446	2046	3046	4046
20	3.3	10	1.0	10.0	12.5	4	4	A	0147	0247	0347	0447	2047	3047	4047
20	[illegible]	20	1.0	10.0	12.5	4	4	A	0148	0248	0348	0448	2048	3048	4048
20	6.8	10	1.2	12.0	15.0	4	4	A	0149	0249	0349	0449	2049	3049	4049
20	10.0	10	4.0	40.0	50.0	6	4	B	0150	0250	0350	0450	2050	3050	4050
20	[illegible]	10	4.0	40.0	50.0	6	4	B	0151	0251	0351	0451	2051	3051	4051
20	22.0	20	4.0	40.0	50.0	6	6	B	0152	0252	0352	0452	2052	3052	4052
20	27.0	10	5.0	50.0	62.5	6	6	B	0153	0253	0353	0453	2053	3053	4053
20	56.0	10	9.0	90.0	112.5	6	6	C	0154	0254	0354	0454	2054	3054	4054
20	68.0	10	10.0	100.0	125.0	6	6	C	0155	0255	0355	0455	2055	3055	4055
20	68.0	20	10.0	100.0	125.0	6	6	C	0156	0256	0356	0456	2056	3056	4056
20	56.0	10	10.0	100.0	125.0	6	6	C	0157	0257	0357	0457	2057	3057	4057
20	100.0	10	15.0	150.0	188.0	6	6	C	0158	0258	0358	0458	2058	3058	4058
20	100.0	20	15.0	150.0	188.0	6	6	C	0159	0259	0359	0459	2059	3059	4059
20	120.0	10	15.0	150.0	188.0	6	8	D	0160	0260	0360	0460	2060	3060	4060
20	150.0	10	20.0	200.0	250.0	6	8	D	0161	0261	0361	0461	2061	3061	4061
20	150.0	20	20.0	200.0	250.0	8	8	D	0162	0262	0362	0462	2062	3062	4062
20	180.0	10	20.0	200.0	250.0	8	8	D	0163	0263	0363	0463	2063	3063	4063

TABLE I. Dash number characteristics and PINs - Continued.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at +25°C	DC leakage at +85°C	DC leakage at +125°C	Dissipation factor at +25°C +85°C	Dissipation factor at -55°C +125°C	Case size	PIN M39003/03- failure rate level for [illegible] 1.0	0.1	0.01	0.001	0.1	0.01	0.001
Volts	µF	percent	µA	µA	µA	percent	percent								
[illegible]	[illegible]	10	3.3	33.0	41.0	6	6	A	0164	0264	0364	0464	2064	3064	4064
	8.2	10	7.5	55.0	44.0	6	6	B	0165	0265	0365	0465	2065	3065	4065
	10.0	10	9.2	55.0	55.0	6	6	B	0166	0266	0366	0466	2066	3066	4066
	[illegible]	20	9.2	92.0	55.0	6	6	B	0167	0267	0367	0467	2067	3067	4067
	[illegible]	10	10.0	100.0	125.0	6	6	[illegible]	0168	0268	0368	0468	2068	3068	4068
	33.0	20	10.0	100.0	125.0	6	6	[illegible]	0169	0269	0369	0469	2069	3069	4069
	39.0	10	10.0	100.0	125.0	6	6	[illegible]	0170	0270	0370	0470	2070	3070	4070
	47.0	10	10.0	100.0	125.0	6	6	[illegible]	0171	0271	0371	0471	2071	3071	4071
	47.0	20	10.0	100.0	125.0	6	6	[illegible]	0172	0272	0372	0472	2072	3072	4072
	56.0	10	15.0	150.0	188.0	6	6	D	0173	0273	0373	0473	2073	3073	4073
	68.0	10	15.0	150.0	188.0	6	6	D	0174	0274	0374	0474	2074	3074	4074
	68.0	20	15.0	150.0	188.0	6	6	D	0175	0275	0375	0475	2075	3075	4075
[illegible]	1.2	10	3.9	9.0	11.0	6	6	A	0176	0276	0376	0476	2076	3076	4076
	[illegible]	10	4.2	12.3	15.0	6	6	A	0177	0277	0377	0477	2077	3077	4077
	1.5	20	4.2	12.3	15.0	6	6	A	0178	0278	0378	0478	2078	3078	4078
	[illegible]	10	4.5	37.0	56.0	6	6	B	0179	0279	0379	0479	2079	3079	4079
	[illegible]	10	4.5	45.0	56.0	6	6	B	0180	0280	0380	0480	2080	3080	4080
	6.8	20	4.5	45.0	56.0	6	6	B	0181	0281	0381	0481	2081	3081	4081
	[illegible]	10	10.0	100.0	125.0	6	6	C	0182	0282	0382	0482	2082	3082	4082
	22.0	20	10.0	100.0	125.0	6	6	C	0183	0283	0383	0483	2083	3083	4083
	[illegible]	10	10.0	100.0	125.0	6	6	C	0184	0284	0384	0484	2084	3084	4084
	[illegible]	10	10.0	100.0	125.0	6	6	D	0185	0285	0385	0485	2085	3085	4085
	[illegible]	20	10.0	100.0	125.0	6	6	D	0186	0286	0386	0486	2086	3086	4086
	[illegible]	10	10.0	100.0	125.0	6	6	D	0187	0287	0387	0487	2087	3087	4087

Reliability data:

Specification for failure rate levels shall be as follows in accordance with MIL-R-39035.

[illegible] for [illegible] shall be as follows:

[illegible]	[illegible]
10 percent	21 percent

CONCLUDING MATERIAL

Custodians:
  Army - ER
  Navy - EC
  Air Force - 85

Preparing activity:
  DLA - ES

(Project 5905-[illegible])

Review activities:
  Army - ER
  Navy - AS, MC, OS, SH
  Air Force - 19, 99, 99